EXECUTIVE SUMMARY

Overtime cost has climbed for six months. Nobody trusts the numbers.

WHY THE NUMBER IS CONTESTED

Attendance looks normal

HR reports that attendance looks normal.

Payroll uses approved OT requests

Payroll pays overtime from approved OT requests, not biometric punches.

HCM views conflict

Different HCM modules produce conflicting views of employee activity and payments.

THE ANSWER

Build one trusted data model.

Reconcile the records. Identify where payroll is leaking.

345,280
Raw punches

12
Source files

5
HCM modules

~1,200
Employees

Scale of the evidence · Raw punches, source files, HCM modules and employees

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DATA FOUNDATION

From messy exports to a warehouse we can prove.

01	02	03	04	05
Raw Exports	**Raw Schema**	**Staging Schema**	**Data Warehouse**	**10 SQL Answers**
12 source files	Loaded untouched; idempotent ingestion	Cleaned, standardized and typed	Star schema with an SCD Type 2 employee dimension	Evidence-based business findings

50 duplicate employee records merged.

16,175 duplicate punches removed.

6,962 unmapped punches isolated in a rejected-records table.

100% reconciliation rule: raw rows = staging rows + rejected rows, for every applicable table.

DEV-SITE-01 ran 1 hour ahead from 1 July 2026. Median check-in gap shifted from approximately 0 to 60 minutes. The correction was applied in code only from 1 July 2026 onward.

MANUAL INSERTION AREA

Warehouse Schema / ER Diagram

Insert the actual Warehouse Schema / ER Diagram here.

Clock correction was implemented in code and applied only from 1 July 2026 onward.

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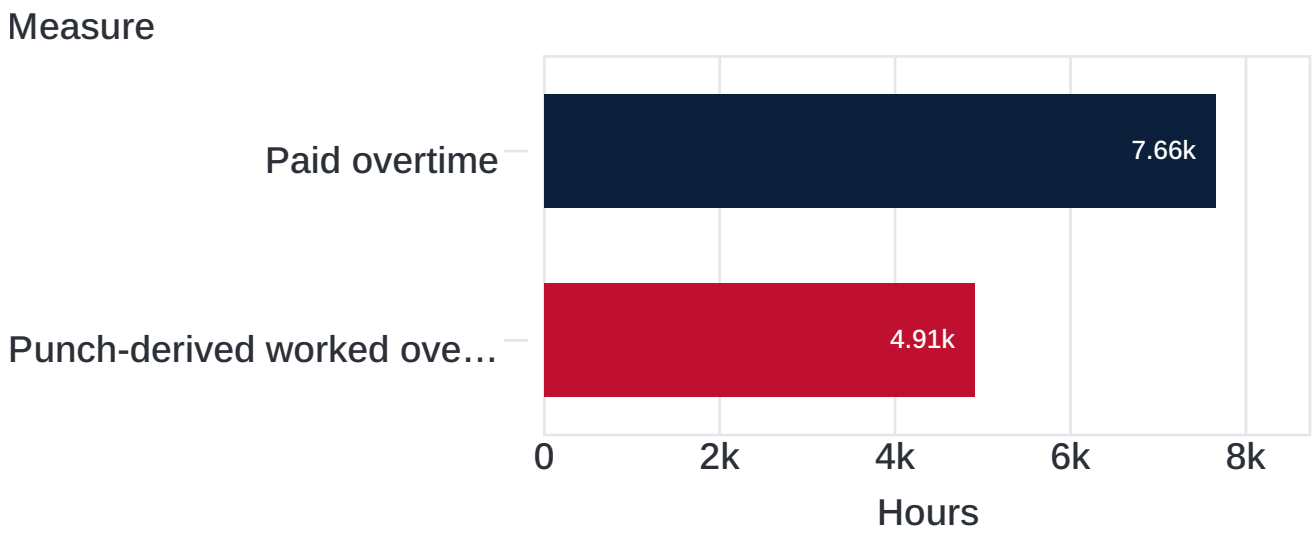
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KEY FINDING · OVERTIME

Faisalabad Stitching is paid 56.1% more overtime than punches support.

PAID VERSUS PUNCH-DERIVED OVERTIME HOURS

Paid overtime materially exceeds punch-derived worked overtime in Faisalabad Stitching.



VARIANCE

+2,753 hours

+56.1% paid above punch-derived worked overtime.

Paid OT exceeded computed OT by more than 15% in 5 of 6 months.

Unique concentration

No other department exceeded this threshold.

Stable computed OT

Overall computed OT remained approximately 6,100–6,500 hours per month.

What the data does not show

The available punch data does not support the conclusion that rising payroll cost is driven simply by employees working more overtime.

Recommended action: cap payable overtime at punch-verified hours, subject to approved exceptions, and audit Faisalabad Stitching OT approvers.

Overtime crept up. It climbed for six months. Nobody trusts the numbers.

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KEY FINDING · PAYROLL CONTROLS

Payroll controls failed: employees were paid without supporting attendance or active employment.

ZERO ATTENDANCE AND ZERO LEAVE

58

employee-months

Paid despite zero recorded attendance and zero leave.

Example: one employee received PKR 85,925 in April despite having no recorded attendance.

PAYMENTS AFTER SEPARATION

PKR 1,948,000

paid to 5 employees after separation

Largest identified case: PKR 660,000 over 5 months for one employee.

These are identified payment exposures requiring validation, not automatically confirmed fraud or guaranteed recoverable losses.

01

Reconcile before release

Reconcile payroll against the HR separation list before every payroll release.

02

Review unsupported pay

Flag zero-attendance, zero-leave employees for mandatory review.

03

Validate and recover

Initiate audit and recovery review for PKR 1,948,000, subject to verification and applicable policy.

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KEY FINDING · LEAVE CONTROL

20 employees exceeded their annual leave entitlement in just six months.

REVIEW POPULATION

20

employees above entitlement

Exceeded the 14-day annual entitlement during the six-month review period.

ENTITLEMENT VERSUS OBSERVED USAGE

Annual entitlement	Worst case observed
14 days	28 days
	14 days over entitlement

What counted

Leave counted only when both manager and HR approved it.

What was excluded

Cancelled leave was excluded.

Control gap

The approval workflow has no leave-balance check.

Recommended actions: enforce an available-balance check; prevent approvals above entitlement unless an authorized exception is recorded; review existing excess leave and apply unpaid leave or other adjustments only where permitted by company policy.

Overtime cost has climbed for six months. Nobody trusts the numbers.

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CONTROL AGENDA

Three control fixes.

Automated checks behind every number.

Every number here comes from a SQL query in the repo.

RECOMMENDED ACTIONS

01 Control Overtime Payments

Tie payable OT to biometric punch-verified hours, starting with Faisalabad Stitching; route exceptions through documented approval.

02 Block Unsupported Payroll

Reconcile payroll against separation records and attendance before each release; flag exceptions for review.

03 Enforce Leave Entitlement

Add an automated leave-balance check to the manager and HR approval workflow.

DATA QUALITY AND REPRODUCIBILITY

- Automated data quality checks passed; results stored in the dq_results table.
- Primary keys are unique.
- 0 foreign-key violations.
- 0 overlapping SCD Type 2 employee-dimension ranges.
- Raw row counts reconcile to staging plus rejected records for every applicable table.
- The complete pipeline can be rebuilt with one command.

REBUILD COMMAND

```
python
run_pipeline.py
```

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